Data Analysis with Python

Cheat Sheet: Model Development

Process	Description	Code Example
Linear Regression	Create a Linear Regression model object	`from sklearn.linear_model import LinearRegression` `lr = LinearRegression()`

Train Linear Regression model	Train the Linear Regression model on decided data, separating Input and Output attributes. When there is single attribute in input, then it is simple linear regression. When there are multiple attributes, it is multiple linear regression.	`X = df[[‘attribute_1’, ‘attribute_2’, ...]]`
		`Y = df['target_attribute']`
		`lr.fit(X,Y)`

Generate output predictions	Predict the output for a set of Input attribute values.	`Y_hat = lr.predict(X)`
Identify the coefficient and intercept	Identify the slope coefficient and intercept values of the linear regression model defined by $\hat{y} = mx + c$ Where m is the slope coefficient and c is the intercept.	`coeff = lr.coef` `intercept = lr.intercept_`
Residual Plot	This function will regress y on x (possibly as a robust or polynomial regression) and then draw a scatterplot of the residuals.	`import seaborn as sns` `sns.residplot(x=df['attribute_1'], y=df['attribute_2'])`

Distribution Plot	This function can be used to plot the distribution of data w.r.t. a given attribute.	`import seaborn as sns`
		`sns.distplot(df['attribute_name'], hist=False)`
		`# can include other parameters like color, label and so on.`

Polynomial Regression	Available under the numpy package, for single variable feature creation and model fitting.	`f = np.polyfit(x, y, n)`
		`#creates the polynomial features of order n`
		`p = np.poly1d(f)`
		`#p becomes the polynomial model used to generate the predicted output`
		`Y_hat = p(x)`
		`# Y_hat is the predicted output`